Measuring Observable Web Dependency Exposure under International Connectivity Disruption: A Request-Level Analysis of Websites Used in Taiwan

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***Abstract*—Taiwan is a highly connected island economy whose international Internet connectivity depends heavily on submarine cables. However, understanding how cable disruptions may affect commonly used web services requires more than infrastructure-level indicators. This paper examines observable web-service dependency exposure under severe degradation or loss of international connectivity associated with submarine cable disruptions, from a user-centric perspective. We propose a request-level measurement framework that analyzes resource requests generated during homepage loading.**

We processed 2,509 target entries covering websites commonly used in Taiwan, including both domestic and global services, and successfully measured 2,179 unique sites. Of these, 856 sites (39.3%) were classified as foreign-dependent, indicating direct exposure under degraded international connectivity scenarios. Another 1,080 sites (49.6%) had no observed foreign resource but relied on local endpoints of multinational cloud and CDN providers, whose operational continuity under severe international connectivity disruption remains uncertain. Only 243 sites (11.2%) were locally contained at the observed frontend layer. Together, 88.8% exhibited either direct foreign or cloud-local dependency exposure.

These findings reveal that modern web services exhibit significant hidden dependencies on global infrastructure, even when resources appear locally served. They highlight the importance of examining application-layer dependencies alongside infrastructure-level indicators when assessing the potential impact of submarine cable disruptions. This work contributes a scalable and reproducible methodology for quantifying observable web dependency, comparing dependency patterns, tracking changes in dependency exposure over time, and examining how large-scale connectivity disruption may affect modern web ecosystems.

***Index Terms*—observable web dependency, Internet measurement, submarine cables, content delivery networks, cloud infrastructure, network resilience**

I. Introduction

International submarine cables are critical infrastructure for global Internet connectivity, particularly for highly digitized island economies whose external traffic depends heavily on a limited number of cable systems. Taiwan is such a case: its overall Internet usage reached 88.75% in 2024 [1], yet the island's global Internet connectivity relies primarily on only 16 international submarine cable systems [2].

In such a highly digitalized setting, large-scale submarine cable disruptions raise concerns about whether the web services people rely on can remain reachable and operational amid international connectivity degradation.

Existing studies of submarine cable resilience primarily focus on infrastructure redundancy, routing diversity, repair capacity, and network-level resilience metrics [3]–[6]. Prior research on web dependency has also shown that modern websites commonly rely on third-party infrastructure and public cloud ecosystems [7]–[9]. However, relatively little work directly evaluates observable application-layer dependency exposure under large-scale international connectivity disruption scenarios from a user and country-level perspective.

This paper investigates the problem through request-level measurement of websites commonly used in Taiwan. We analyze the resource requests generated during webpage loading to identify observable dependency exposure on foreign resources and multinational cloud and CDN infrastructure. Although these endpoints may be located on the island, their local delivery may still depend on foreign origins or provider systems, leaving availability uncertain in the event of international connectivity disruption.

This paper makes three contributions:

- We propose a scalable and reproducible request-level measurement framework for analyzing web dependency exposure in island economies under international connectivity degradation scenarios.
- We apply the framework to 2,179 websites commonly used in Taiwan and classify observed dependencies into foreign-dependent, cloud-dependent, and locally-contained categories.
- We quantify dependency exposure across the measured dataset and show that 88.8% of websites exhibit either direct foreign dependency exposure or reliance on local endpoints of multinational cloud and CDN providers.

II. Background and Related Work

A. Submarine Cable and Internet Resilience

Internet resilience has been studied across physical, logical, and application layers. Prior work on routing resilience shows that physical connectivity alone does not guarantee reachability, because routing policies, failures, and path selection can affect whether traffic is delivered successfully [3], [4]. Policy-oriented studies of submarine cable resilience emphasize redundancy, repair capacity, geographic diversity, and international cooperation [5], [10]. Country-level indices, such as the Internet Resilience Index, aggregate indicators of infrastructure, performance, security, and markets to describe national Internet ecosystems [6].

These approaches help establish that submarine cable disruption is a serious systemic risk, but they provide limited visibility into how serious the impact may be for user-facing services, which services are exposed, and how progress in mitigation can be measured over time. Answering these questions requires moving beyond network-level readiness to examine application-layer dependency structures that are visible from the user side.

B. Third-Party and Cloud Dependencies

Modern web services commonly load observable third-party resources, including CDN-hosted assets, analytics scripts, web fonts, JavaScript libraries, authentication-related endpoints, and APIs. Many of these resources may be served from public cloud platforms. Prior studies have shown that third-party dependencies are widespread and can create concentration risks and indirect failure propagation [7]–[9]. Web dependency analysis has also been used to identify resource dependencies and their effect on rendering [11].

Our work builds on this line of research but frames the measurement around a specific disruption scenario: large-scale international connectivity degradation or disruption. We therefore distinguish between foreign resource exposure and domestic endpoints operated by multinational cloud and CDN providers. The latter may appear local at the network edge, but their continuity under disconnection can depend on cache state, origin placement, provider control planes, and authentication mechanisms.

III. Problem Definition

We consider a scenario in which Taiwan's international Internet connectivity is severely degraded, unstable, or partially interrupted due to large-scale submarine cable disruptions. The central question is how prevalent observable foreign and cloud dependencies are among websites commonly used in Taiwan, how concentrated cloud-local dependencies are among provider ecosystems, and what these dependencies imply for initial service access.

We use a programmatic browser to observe the distribution of resource requests generated while loading each homepage. A homepage is the first user-visible point of interaction with a service and includes some of the frontend resources required for basic rendering and interaction. We treat this observation as a scalable proxy indicator for comparing observable dependency exposure across a large number of websites. However, it does not capture the complete backend architecture, cloud control-plane dependencies, or full user workflows. Although these results cannot confirm whether a service would remain available during a disruption, they provide a preliminary indicator of its potential availability.

We focus on three research questions:

- **RQ1:** What fraction of commonly used websites expose foreign resource dependencies on the homepage?
- **RQ2:** What patterns of reliance on local endpoints of multinational cloud and CDN providers are observed among websites commonly used in Taiwan?
- **RQ3:** Do government and education websites exhibit different foreign-dependency rates from the overall dataset?

We first classify individual retained homepage resource hostnames by endpoint location and provider. A *foreign resource* has an endpoint classified as outside Taiwan; its delivery may be affected by severe degradation or loss of international connectivity. A *cloud-local resource* has an endpoint classified as local whose ASN matches a multinational cloud or CDN provider in our curated registry. This indicates local delivery in the measurement, but continued service operation during international disconnection may depend on provider or service systems outside Taiwan.

We then aggregate these resource-level classifications to place each website in one of three categories:

- **Foreign-dependent:** at least one foreign resource is observed.
- **Cloud-dependent:** no foreign resource is observed, but at least one cloud-local resource is observed.
- **Locally-contained:** neither foreign nor cloud-local resources are observed among retained homepage resources.

These categories describe observable dependency exposure, not complete service availability. A locally-contained homepage may still depend on foreign backend systems, such as databases, internal APIs, or authentication services, and a cloud-dependent website may remain available if the necessary resources are cached and remain locally operational.

IV. Methodology

A. Dataset

We constructed a list of websites commonly used in Taiwan from ranking snapshots collected on July 20, 2026. The inputs comprised 2,510 Tranco entries ending in `.tw` [12], the Taiwan top 100 from Cloudflare Radar [13], the Taiwan top 100 from AhrefsTop [14], the Taiwan top 50 from SimilarWeb [15], and the Taiwan top 100 from Semrush [16]. We lowercased hostnames, removed a leading `www.`, preserved other subdomains, and merged duplicate hostnames while retaining their source-specific ranks. This produced 2,467 automatically ranked sites. A manually curated list added 42 community and digital-resilience cases; two overlapped with the ranked list,

yielding 2,509 target entries and 2,507 unique target sites. Of these, 2,179 sites completed successfully (86.9%). The measurements were collected on July 21, 2026.

The dataset is intentionally defined as websites used in Taiwan, rather than only websites operated in Taiwan. This includes domestic websites as well as international services commonly accessed by local users. The measurement target is the web homepage; mobile applications are outside the scope of this paper.

B. Measurement Environment

Measurements were conducted from two residential sites in the Taipei metropolitan area using Chunghwa Telecom 500/500 Mbps fixed broadband and the recursive DNS resolver 168.95.1.1.

C. Measurement Pipeline

We loaded each homepage in a programmable Chromium browser and collected its network requests. Requests with `blob:` or unparsable URLs were discarded. The campaign used LowTechFilter hosts ABP version 2026.0720.1 [17] and an AdGuard DNS Filter snapshot last modified on July 20, 2026 [18]; the parser extracted 158,521 distinct domain rules across the two lists. Matching domains were excluded, except that the target hostname and hostnames in a parent or child relationship with it were always retained to avoid filtering the website's own resources. We also excluded `fonts.gstatic.com`, the sole manually excluded hostname, because failures of web-font delivery are less likely to affect core website functionality than failures of scripts, APIs, or content resources. After filtering, requests were deduplicated by hostname within each website. The resulting unit of observation was a unique website-hostname pair, so multiple requests from the same website to the same hostname were counted once. Each observation was then resolved for location and provider classification.

D. Location and Provider Classification

For each observed hostname, we used IPinfo to obtain geolocation and autonomous system information for the first IPv4 address returned by the separate DNS lookup [19]. An address whose IPinfo location was Taiwan was classified as local. To identify multinational cloud and CDN resources, we matched reported ASNs against a 30-provider registry (version 0.1.0, updated January 5, 2026). Taiwan-only providers were excluded. The registry identifies six providers with Taiwan edge nodes.

For endpoints belonging to these providers that IPinfo located outside Taiwan, we applied a location-classification cascade to detect local edge delivery. We first inspected provider-specific response headers (`cf-ray`, `x-amz-cf-pop`, `x-served-by`, `x-azure-ref`, and `x-msedge-ref`) for the Taipei marker `TPE`. If no header indicated Taipei, we queried the LACeS Anycast Census API [20]; an endpoint was reclassified as local when LACeS returned a Taiwan location with `confident` confidence. Otherwise, we sent five ICMP echo requests and reclassified the endpoint as local only when its minimum RTT was below 15 ms. All remaining endpoints were classified as foreign.

Each observation was assigned to one of four classes: domestic-cloud, domestic-direct, foreign-cloud, or foreign-direct. In these class labels, cloud includes multinational cloud and CDN providers identified through ASN matching against our curated registry. A website was then classified by aggregating the resource classes observed during homepage load: websites with any foreign-cloud or foreign-direct resources were classified as foreign-dependent; websites without foreign resources but with at least one domestic-cloud resource were classified as cloud-dependent; and websites with neither foreign resources nor domestic-cloud resources were classified as locally-contained.

TABLE I
COVERAGE OF THE RTT FALLBACK STAGE.

Metric	Count	Share
Measured websites	2,179	–
Websites with RTT fallback	1,243	57.0%
Classification observations	19,046	–
Fallback observations	3,640	19.1%
RTT replies	3,064	84.2%
$<$15 ms	2,394	78.1%
$\geq$15 ms	670	21.9%

E. RTT Fallback Coverage and Sensitivity

RTT is the final stage of the location-classification cascade and is applied only when IPinfo, provider-specific response headers, and LACeS cannot establish the location of a target cloud or CDN endpoint. Across the 2,179 successfully measured websites, the browser recorded 262,926 raw HTTP requests. Filtering and within-site hostname deduplication produced 19,046 observations for classification. Of these, 3,640 observations from 976 distinct hostnames across 1,243 websites entered the RTT fallback stage. Numeric RTT values were obtained for 3,064 observations; 2,394 were below 15 ms and 670 were at least 15 ms. Table I summarizes this coverage. The percentages use the following denominators: 2,179 websites for websites with RTT fallback, 19,046 observations for entry into RTT fallback, 3,640 fallback observations for successful measurements, and 3,064 successful measurements for the two threshold outcomes.

The successful measurements exhibit a bimodal distribution (Fig. 1). Only 130 observations, or 4.2%, fall in the 10–30 ms transition range. We therefore place the 15 ms cutoff within this sparse interval and classify an endpoint as local only when its minimum RTT is less than 15 ms. This conservative choice keeps an ambiguous endpoint classified as foreign.

We also recomputed website classifications at 10 ms and 20 ms. Relative to the 15 ms baseline, a 10 ms threshold moves 27 websites (1.2%) from cloud-dependent to foreign-dependent. A 20 ms threshold moves five websites (0.2%) in the opposite direction. The locally-contained count is unchanged because RTT is used only to locate resources already identified as be-

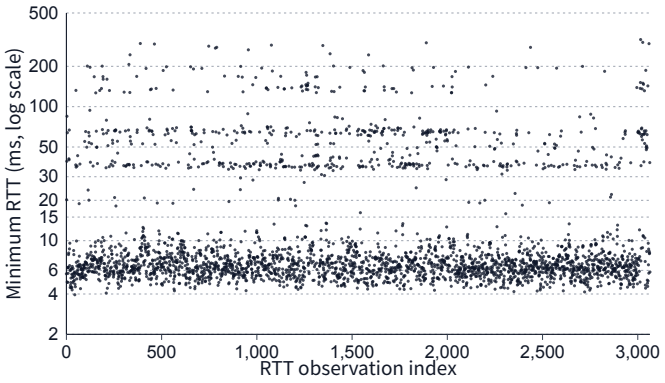


Fig. 1. Distribution of 3,064 successful minimum-RTT measurements.

TABLE II
WEBSITE CLASSIFICATION SENSITIVITY TO THE RTT THRESHOLD.

RTT	Foreign	Cloud	Local	Changed
10 ms	883	1,053	243	27 (1.2%)
15 ms	856	1,080	243	–
20 ms	851	1,085	243	5 (0.2%)

longing to the selected multinational cloud or CDN providers. Table II shows that the main findings are stable across these alternatives.

F. Limitations of the Method

The measurement captures observable frontend resource dependencies, not the complete internal architecture of a service. It does not evaluate application workflows such as login, search, checkout, or content submission. It also does not inject failures or emulate a full international disconnection. Therefore, the results should be interpreted as indicators of observable dependency exposure during homepage loading, not as a direct measure of complete service availability or actual outage behavior.

A website is classified as foreign-dependent if at least one retained resource hostname is classified as foreign. Although the ad-blocking hostname lists filter likely non-essential requests, the lists may be incomplete. Thus, a site classified as foreign-dependent may still load during a connectivity disruption.

The 15 ms RTT cutoff remains a fallback heuristic rather than physical proof of location; routing changes, congestion, and nearby foreign nodes can affect an individual result even though the threshold sensitivity analysis shows limited aggregate impact.

The dataset is based on websites commonly used in Taiwan. It is not systematically classified by service sector or weighted by societal criticality. The government and education subsets provide domain-based comparisons, but they are not a substitute for a formal framework for classifying service criticality. Consequently, website counts measure the prevalence of exposure, not the severity of societal impact. This scope allows the method to scale across thousands of websites while providing a consistent initial view of dependency structure.

V. RESULTS

A. Overall Dependency Categories

Fig. 2 shows the overall classification of 2,179 successfully measured websites. We find that 856 websites, or 39.3%, are foreign-dependent. These websites have at least one retained homepage resource hostname whose endpoint was classified as foreign, indicating direct exposure of part of the homepage's resource delivery to international connectivity disruption.

Another 1,080 websites, or 49.6%, are cloud-dependent. They do not show direct foreign resource exposure in our measurement, but they rely on local endpoints of multinational cloud and CDN providers. Their continued operation under prolonged degradation of international connectivity remains uncertain and depends on provider-specific implementation details such as cache persistence, origin placement, and control-plane reachability. Only 243 websites, or 11.2%, are locally-contained at the observed frontend resource layer.

Foreign-dependent and cloud-dependent websites together account for 88.8% of the measured dataset.

B. Cloud and CDN Provider Concentration

Local cloud and CDN endpoints are common among the 1,323 websites without observed foreign resources. Table III reports, for each provider, the number and percentage of websites using at least one local endpoint. Since a website may request resources from multiple providers' local endpoints, the percentages are not mutually exclusive. Google is the most common, appearing on 965 websites. Cloudflare appears on 480 websites, followed by Amazon, Akamai, Microsoft, and Fastly.

The location mix also varies substantially by provider. Table IV reports the domestic and foreign observations associated with each of the six providers that operate edge nodes in Taiwan. Google has 7,393 domestic and 63 foreign

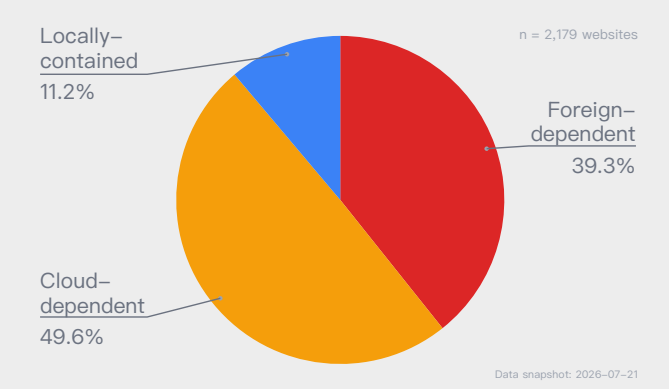


Fig. 2. Overall dependency classification of 2,179 successfully measured websites.

TABLE III
USE OF LOCAL CLOUD OR CDN ENDPOINTS AMONG 1,323 WEBSITES WITHOUT OBSERVED FOREIGN RESOURCES. PROVIDERS ARE NON-EXCLUSIVE.

Provider	Websites	Usage Rate
Google	965	72.9%
Cloudflare	480	36.3%
Amazon	138	10.4%
Akamai	104	7.9%
Microsoft	38	2.9%
Fastly	4	0.3%

TABLE IV
PROVIDER OBSERVATIONS BY ENDPOINT LOCATION.

Provider	Domestic	Foreign
Google	7,393	63
Cloudflare	3,051	21
Amazon	1,382	522
Akamai	446	13
Fastly	6	369
Microsoft	196	143

observations, whereas Fastly has six domestic and 369 foreign observations.

This concentration matters because edge localization does not ensure resilience if international connectivity is degraded or lost. Cached static resources may remain available temporarily, while dynamic content, cache misses, identity services, or management-plane dependencies may fail. The continued operation of a small set of major public cloud and CDN providers therefore becomes a key factor in the resilience of many websites.

C. Global Cloud and Other Providers

For this comparison, *global cloud* includes the selected multinational cloud and CDN providers, while *other providers* comprises the remainder. Across the 2,179 websites successfully measured, 87.7% rely on at least one global-cloud resource. More specifically, 86.3% rely on domestic global-cloud endpoints and 34.6% rely on foreign global-cloud endpoints. Table V summarizes this breakdown by resource location and provider group.

Only 39 websites, or 1.8% of all measured websites, use only foreign resources without any domestic resources. This suggests that CDN and cloud localization already provide substantial domestic delivery for most websites. However, this localization should be treated as a resilience opportunity that requires validation, not as proof of availability under disconnection.

D. Resource Source Distribution

Among the 18,969 observations with provider information from IPinfo, Google accounts for 39.7%, followed by Cloudflare at 16.4%, Amazon at 10.4%, Chunghwa Telecom, a major domestic telecommunications provider, at 8.7%, and Facebook at 7.7%. This concentration indicates that a relatively small number of providers influence the application-layer resilience of a large fraction of the measured web. Table VI summarizes the top providers by observation share.

E. Government and Education Websites

Government and education websites show lower direct foreign dependency exposure than the overall dataset. We identify these groups by domain suffix: 235 websites under .gov.tw and 255 websites under .edu.tw were included in the measured dataset.

Sixteen government websites (6.8%) and 34 education websites (13.3%) exhibited foreign resource dependency. Both shares are far below the overall foreign-dependent share of 39.3%. This difference may reflect greater use of domestic hosting or institutional networks. Table VII summarizes this breakdown by group.

VI. DISCUSSION

A. From Connectivity to Service Dependency

The measurements complement infrastructure-level indicators such as cable counts, bandwidth capacity, routing diversity, and physical topology by revealing the resource dependencies encountered when users load websites. These dependencies connect network disruption scenarios to the services people use: infrastructure indicators describe connectivity, while application-layer measurements identify foreign resources and cloud-local delivery that warrant service-continuity assessment.

Direct foreign-resource exposure is observed for 39.3% of measured websites. Another 49.6% have no observed foreign resource but use local endpoints of selected multinational cloud and CDN providers. Thus, 88.8% fall into one of these two observable exposure categories. Considering only direct foreign resources would leave the cloud-local dependencies of nearly half the measured websites outside the assessment. Separating this group makes its reliance on multinational providers visible while leaving continuity during disconnection as an open operational question.

The observed dependency patterns extend beyond fully foreign-hosted services to include foreign resources and local endpoints of multinational cloud and CDN providers. As a result, resilience assessment should not focus solely on whether a service appears locally hosted. Instead, it should examine whether critical service functions can continue operating under severe degradation of international connectivity.

B. Cloud Edge Localization as Opportunity and Risk

The high prevalence of domestic cloud endpoints presents both an opportunity and a risk. On the one hand, it shows that modern CDN and cloud infrastructure already localize a large share of resources, which could improve performance and provide a basis for continuity during partial disruptions. On the other hand, edge localization can mask deeper dependencies on foreign resources and make service continuity harder to assess from the outside. In such cases, availability may depend on provider-specific capacity, operational policies, and resilience mechanisms that are not externally visible or verifiable.

TABLE V
WEBSITES USING AT LEAST ONE OBSERVED RESOURCE, BY ENDPOINT LOCATION AND PROVIDER GROUP. CATEGORIES ARE NON-EXCLUSIVE.

Type	Domestic	Foreign	Any
Global cloud	1,881 (86.3%)	754 (34.6%)	1,910 (87.7%)
Other providers	1,623 (74.5%)	245 (11.2%)	1,709 (78.4%)
Any resource	2,140 (98.2%)	856 (39.3%)	2,179 (100.0%)

Architecturally, local delivery must be distinguished from the dependencies needed to sustain it. A localized data plane may continue serving cached assets using existing configuration even when a provider's control plane is unreachable. However, cache misses, revalidation, and dynamic requests may require access to foreign origins or application backends. Expiring sessions may require reachable identity services, while configuration changes or certificate renewal may depend on provider management systems. If the required systems are outside Taiwan and become unreachable, these operations may be impaired even while some cached content remains available. The outcome depends on provider architecture, service configuration, and the duration of disconnection; these are possible mechanisms, not failure modes measured in this study. A locally reachable endpoint is therefore evidence of local delivery, not proof that the service can operate autonomously.

Governments and highly digitized island economies should therefore include cloud dependencies in resilience planning. One question is how long cached resources remain available when international connectivity is lost. Another is whether local points of presence can operate autonomously. Assessments should also examine certificate and identity systems during disconnection and the prioritization of critical services under constrained capacity.

C. Implications for Policy and Operations

The findings suggest that critical digital infrastructure evaluation should include application-layer dependency analysis alongside physical connectivity metrics. Operators of critical services should minimize avoidable foreign dependencies and establish tested local fallback mechanisms. Governments should also agree with major cloud and CDN providers on continuity requirements and consider conducting periodic network-disruption exercises.

For service operators, practical improvements include hosting critical frontend assets locally or providing local fallbacks, reducing unnecessary third-party dependencies, and separating essential web libraries from analytics or advertising resources. For high-impact services, homepage availability is not sufficient; essential user workflows should also be tested under constrained or unavailable international connectivity and designed to operate in degraded network environments.

TABLE VI
TOP PROVIDERS AMONG 18,969 OBSERVATIONS WITH PROVIDER DATA.

Provider	Observations	Percentage
Google	7,525	39.7%
Cloudflare	3,109	16.4%
Amazon	1,979	10.4%
Chunghwa Telecom	1,645	8.7%
Facebook	1,460	7.7%
Akamai	518	2.7%
Fastly	375	2.0%

TABLE VII
FOREIGN DEPENDENCY EXPOSURE BY WEBSITE GROUP.

Group	Tested	Foreign	%
Government	235	16	6.8%
Education	255	34	13.3%
All websites	2,179	856	39.3%

VII. CONCLUSION

This paper presents a request-level measurement of observable web dependency exposure under international connectivity disruption scenarios, using websites commonly used in Taiwan as a case study. Of the 2,179 measured sites, 39.3% had at least one retained homepage resource hostname classified as foreign, 49.6% used local endpoints of selected multinational cloud or CDN providers without observed foreign resources, and 11.2% showed neither form of exposure at the observed frontend layer. These findings highlight why submarine cable disruption is also a service-dependency problem: assessing potential service impact requires examining both foreign resource delivery and the continuity of locally served cloud resources, rather than only the location of a website's main host.

By making these dependencies observable at the request level, this work provides a practical basis for measuring service exposure, comparing dependency patterns across website categories, and tracking changes in dependency exposure over time. Future work will classify websites by service sector, such as government, finance, communications, and e-commerce, to support comparisons across different social and economic functions. It will also extend measurements to additional pages and complete interactive workflows, such as login, search, and checkout, and apply the method in other countries to compare dependency patterns.

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